

Course Code	
Course Name	Web Development using Asp.net WebForms, Asp.Net MVC, Asp.net MVC Core & Web API

Duration (in days)	12	Proficiency Level	Fundamental
Pre-requisites	- C# Programming - ADO.NET - ADO.NET Entity Framework Web Development	Target Audience	Campus Hires

Learning Outcome

At the end of the program, participants will be able to learn:

- ASP.NET Architecture
- Data Binding Part
- State Management
- Caching
- Error Handling
- Tracing
- ASP.NET MVC
- Asp.Net Core
- WebAPI and Swagger

Day-wise Session Plan

Day	Unit	Objective(s)	Hours
1	ASP.NET Architecture	• Introduction • ASP.NET Architecture • Introduction to IIS • Dynamic content • Server-side controls • Data binding • Declarative data binding • Code behind • Page lifecycle • Event handling	3
	Control-based Programming	• HtmlControls • WebControls • Control state management • Implications of state retention • ViewState • Decoding ViewState • Disabling ViewState • Change notification events and ViewState • EventTarget and EventArgs fields • ControlState • Explicit use of ViewState	3
	User Defined Controls	• Web User Controls and Custom Controls	2

2	User Interface Elements	• Templated sites • Master page basics • Creating master pages • Master page control generation • Working with master pages • Nested master pages • Navigation controls • Summary	4
	Configuration Architecture	• Machine.config file • Web.Config • Application level web.config • HttpModules and HttpHandlers • Global.asax file and Application level events	2
	Data Binding Part	• Databinding in ASP.NET • Page-level binding • Binding text and values • How declarative data binding works • SqlDataSource • GridView • DetailsView • Data Templates • DataList	2
3	Introduction to ASP.NET MVC	• Introduction • History of Web Forms • Design Goals • The MVC Design Pattern • Web Forms versus MVC • Routing • Controllers • Conventions • Views and Typed Views • View Helpers	3
	ASP.NET MVC Controllers	• Routes and Controllers • Request Processing • Actions • Selector Filters • Action Parameters • Action Results • Custom Action Results	3
	ASP.NET MVC Views	• Razor View Engine • View Content • View Data • ViewBag • TempData • Partial Views • Validation • Alternate View Engines	2
	HTML Helpers	• Introduction to HTML helpers • DisplayNameFor() • DisplayFor() • BeginForm() • LabelFor()	2

4		• EditorFor() • ValidationMessageFor() • RadioButtonFor() • DropDownListFor() • ListBoxFor() • CheckBoxFor() • AntiForgeryToken()	
	ASP.NET MVC Models	• Introduction • What is the Model? • Model Conventions • Building a Simple Model • Model Binders • Edit the Model • Using the Form Collection • Including and Excluding Properties • Using Parameters • Model Parameters • Custom Model Binder	2
	Exception handling	• Handling Exceptions • ELMAH • Logging Exceptions • Global Exceptions	2
	Filters	• Action Filters • Result Filters • Exception Filters • Auth Filters • [OutputCache] • [HandleError] • [NonAction] • [ActionName] • Custom Action Filters	2
5	Validations	• Client Side Validation • Server Side Validations • Validating Models • Data Annotations	4
	Database Programming	• Connecting to D/Bs using Ado.Net / EF • CRUD operations • Repository Pattern	4
6	Routing	• RouteTable • Attribute Routing • Conventional Routing • URL Routing • Parameters in URL • Default Parameter Values • Parameters with Constraints • Literals in URL • Route Constraints • Custom Route Constraints	5
	Asp.Net Core	• Asp.Net Core Intro • Creating Asp.Net Core apps from command line • Running Applications	3
	Startup	• App startup in ASP.NET Core	2

7 & 8		• Launch Settings • wwwroot and appsettings • Program File	
	Middleware	• ASP.NET Core Middleware • UseCors • UseAuthentication • UseAuthorization • UseStaticFiles • MapRazorPages • UseExceptionHandler • Custom Middlewares	4
	Hosting	• .NET Generic Host in ASP.NET Core • ASP.NET Core Web Host • OWIN • Kestrel	4
	Asp.Net MVC Core	• Asp.Net MVC Core • Folder structure • Changes in MVC Core	4
	Logging	• Introduction to Logging • ILogger • Logging Configuration • Logging Providers • ILogger in Controller • HTTP Logging • HTTP Logging Options • Serilog Basics	4
9	Ado.Net EF Core	• Ado.Net EF Core Introduction • Ado.Net EF vs Core EF • CRUD Operations using Ado.Net EF Core	8
10	Web API	• Introduction to Web API • Creating Web API Project • Web API Controllers • [HttpPost] • [HttpGet] • IActionResult vs ActionResult • ControllerBase	4
	CRUD operations	• EntityFrameworkCore with Web API • CRUD operations • Postman tool	4
11	Swagger / Open API	• Swagger / Open API - Basics • Documentation Comments • Content Negotiation	2
	Versioning	• Setting up Versioning in .NET Web API • Query String-Based versioning in Web API • URL Based Versioning • HTTP Header-Based Versioning	2
	Sorting,Filtering,Paging	• Introduction • Adding Server Side Sorting • Adding Server Side Filtering • Adding Server Side Paging	4
	Minimal APIS	• Introduction to Minimal API • Creating Basic Minimal API	

12		• GET and POST with Minimal API • Route Parameters • MapGroups • IResult • End Point Filters • IEndpointFilter	8
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